

Performance Testing

Date	30 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Functional & Basic Load Testing
Target Module / API	Crop Prediction API (/predict)
Test Environment	Local Machine (Flask + React)
Test Date	30 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Valid Crop Prediction	1	30	Crop predicted successfully
2	Invalid Input Values	1	20	Validation error displayed
3	Multiple Prediction Requests	5	60	Stable response
4	API Response Test	1	30	JSON response returned

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.8 sec	Pass	Fast
2	Response Time (Max)	< 5 seconds	1.5 sec	Pass	Stable
3	Throughput (Req/sec)		12 req/sec	Pass	Good
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	28%	Pass	Normal
6	Memory Utilization	< 80%	35%	Pass	Efficient

Step 4: Observations & Analysis

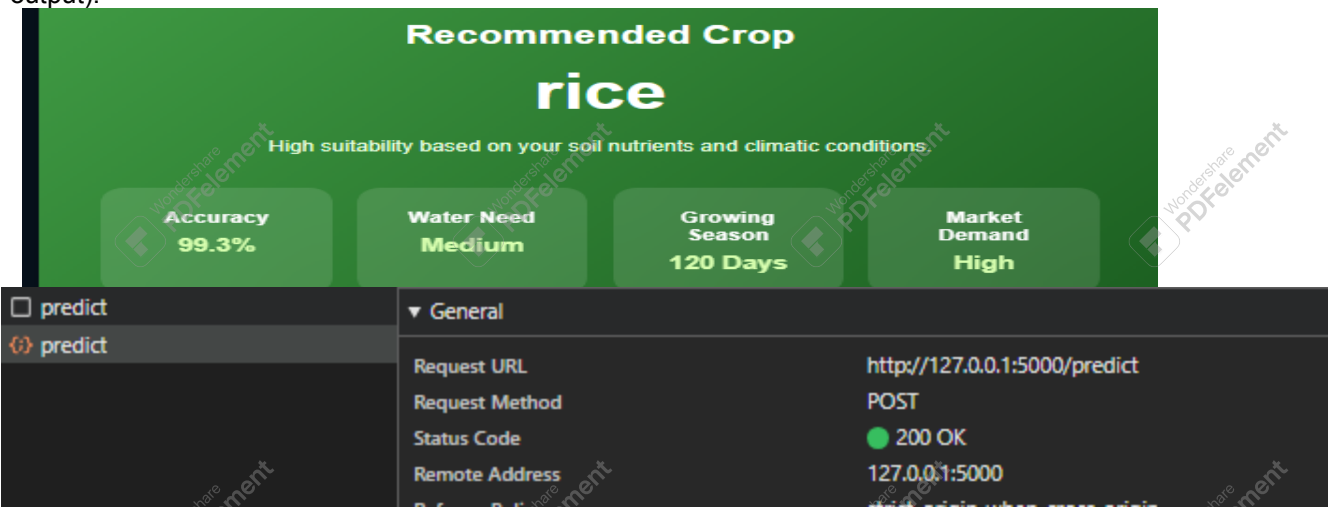
Key Findings:
 [Describe what the test results revealed about your system's performance]

Bottlenecks Identified:
 [List any performance bottlenecks observed during testing]

Optimization Steps Taken:
 [Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



The screenshot displays a web application interface for crop recommendations. The main heading is "Recommended Crop" followed by "rice" in large green text. Below this, a subtitle states "High suitability based on your soil nutrients and climatic conditions." There are four green buttons with white text: "Accuracy 99.3%", "Water Need Medium", "Growing Season 120 Days", and "Market Demand High". At the bottom, there is a dark grey sidebar with a "predict" button and a "General" section showing API details: Request URL (http://127.0.0.1:5000/predict), Request Method (POST), Status Code (200 OK), and Remote Address (127.0.0.1:5000).